

Secure Coding Review Report

CodeAlpha Cyber Security Internship

Task 2: Secure Coding Review

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1. Introduction

Secure coding is the practice of developing software in a way that protects applications from security threats and vulnerabilities. It helps ensure that applications remain reliable, secure, and resistant to cyberattacks.

The purpose of this report is to perform a security review of the **Resume Analysis and Job Matching System** by identifying common security vulnerabilities and recommending secure coding practices that can improve the application's overall security.

2. Application Overview

Application Name

Resume Analysis and Job Matching System

Purpose

The Resume Analysis and Job Matching System is designed to help match job seekers with suitable job opportunities by analyzing resumes and comparing candidate qualifications with job requirements.

Main Features

- Resume submission
- Candidate information management
- Job matching
- Database storage
- User login

Since the application stores personal information such as names, email addresses, phone numbers, educational details, and resumes, protecting this data is essential.

3. Vulnerability 1 – SQL Injection

Description

SQL Injection is one of the most common web application vulnerabilities. It occurs when an application directly inserts user input into SQL queries without proper validation or parameterized queries.

An attacker can manipulate database queries by entering malicious SQL commands into input fields such as login forms or search boxes.

Possible Impact

- Unauthorized access to database records
- Data modification
- Data deletion
- Leakage of sensitive information

Recommendation

- Use parameterized queries (Prepared Statements)
- Validate all user input
- Avoid dynamic SQL queries
- Apply the principle of least privilege to database accounts

4. Vulnerability 2 – Weak Authentication

Description

Authentication is the process of verifying a user's identity before granting access to an application.

Weak authentication occurs when applications use weak passwords, lack account protection, or do not implement additional security measures such as Two-Factor Authentication (2FA).

Possible Impact

- Unauthorized account access

- Account takeover
- Data theft
- Privacy violations

Recommendation

- Enforce strong password policies
- Hash passwords before storing them
- Enable Two-Factor Authentication (2FA)
- Limit login attempts to prevent brute-force attacks

5. Vulnerability 3 – Sensitive Data Exposure

Description

Sensitive data exposure occurs when confidential user information is stored or transmitted without proper protection.

The Resume Analysis and Job Matching System stores personal information including resumes, email addresses, phone numbers, and educational records. If this information is not properly protected, attackers may gain unauthorized access.

Possible Impact

- Identity theft
- Privacy breaches
- Data leakage
- Loss of user trust

Recommendation

- Encrypt sensitive data
- Use HTTPS for secure communication
- Restrict database access
- Perform regular security backups

6. Secure Coding Best Practices

To improve application security, developers should follow secure coding practices throughout the software development process.

Input Validation

Validate all user input before processing it to prevent malicious data from entering the system.

Parameterized Queries

Always use prepared statements instead of directly embedding user input into SQL queries.

Password Security

Store passwords using secure hashing algorithms and encourage users to create strong passwords.

Data Encryption

Encrypt confidential information while storing it and during transmission.

Error Handling

Display user-friendly error messages without revealing sensitive system information.

Regular Security Testing

Perform regular security reviews, vulnerability assessments, and software updates.

7. Conclusion

This security review identified three major security risks that could affect the Resume Analysis and Job Matching System:

- SQL Injection
- Weak Authentication
- Sensitive Data Exposure

Implementing secure coding practices such as parameterized queries, strong authentication mechanisms, encryption, and regular security testing can significantly improve the application's security and protect user data from common cyber threats.

8. References

1. OWASP Top 10 Web Application Security Risks
2. OWASP Secure Coding Practices
3. Microsoft Secure Coding Guidelines
4. NIST Cybersecurity Framework
5. CWE (Common Weakness Enumeration)